Simulation-Based Evaluation of a Kalman-Filter State-Estimation Module for a Primary Crusher Feed Hopper Under Measurement Noise and Model Mismatch

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***Abstract*—State estimation is a core function of process digital twins, but it is often evaluated on data generated by the same model the estimator assumes, which makes the estimator optimal by construction. We evaluate a two-state linear Kalman filter (KF) that reconstructs the ore volume and the unmeasured extraction rate of a primary crusher feed hopper from a noisy level measurement and the known truck dumping flow. The simulated process deliberately departs from the filter model (physical limits, extraction saturation when the hopper empties, mean-reverting ore hardness, random truck arrivals), and the sensor can exhibit colored noise, noise bursts and drift. Over 100 paired Monte Carlo runs per condition, the KF reduces the volume RMSE by 74.7 % with respect to the raw measurement. Against a window-based local regression that uses the same information and is tuned with ground truth, the mean difference is not significant (-3.4 %), yet the KF wins 72 % of the runs. An exploratory stratification locates its losses in runs where the hopper empties, and an ablation with the same seeds shows that extraction saturation causes them: without it, the KF is better in those same runs ($+9.6$ %, 97 % won). The normalized innovation squared (NIS) flags noise bursts and the innovation autocorrelation flags colored noise, but a positive autocorrelation alone does not mean the gain is too low: with colored noise, raising the gain from its nominal value did not appreciably reduce the error, and what separated this case from a low gain was an autocorrelation that barely responded to the tuning. Neither diagnostic flags drift, which is unobservable under the filter model. All results are simulation-based and reproducible bit for bit.**

***Index Terms*—Kalman filter, state estimation, estimator consistency, digital twin, mining, primary crushing, Monte Carlo simulation.**

I. Introduction

In open-pit mining, haul trucks dump ore into the feed hopper (dump pocket) of the primary crusher, which extracts it continuously. Non-contact level sensors used on such hoppers are affected by dust and wall echoes [1], and we assume the extraction rate is not measured at the hopper. A recent systematic review of digital twins (DTs) in mining identifies data quality as a recurring challenge and notes that many proposals remain confined to simulation [2]; in mineral processing, observing data through mass balances before using them for control is a long-standing principle [3], [4].

The Kalman filter [5] is the natural tool to estimate the hopper state from a mass balance and noisy measurements, and state estimation is recognized as a DT function [6], [7]. However, the filter is optimal only when the process matches its model [8]. A study that generates data with the filter's own model can only confirm that the filter works when its assumptions hold. We instead ask how much of the filter's accuracy and consistency survives when they do not, relative to which alternatives, and whether the loss can be detected without ground truth.

Our contributions are: (i) an open, verified estimation module and an evaluation protocol in which the simulated process differs from the filter model in controlled ways; (ii) a comparison against a window-based estimator using the same information, reported per run and not only as a mean, in which an ablation shows that extraction saturation during emptying causes the filter's losses and a calibrated tuning largely compensates them; (iii) a map of which sensor mismatches the in-plant diagnostics can detect, showing that a positive innovation autocorrelation indicated a low gain only when it responded to the tuning; and (iv) the use of the classical closed-form steady-state solution to predict the simulated behavior. We do not propose a new filter, we do not build a full DT (in the taxonomy of [9] as summarized in [10], the module is a digital model of a component), and we do not claim plant performance.

II. Related Work

Digital twins in minerals. Qu *et al.* [11] argue that a DT proper requires real-time bidirectional data flows and that the term is overused; Sepasgozar [12] distinguishes digital shadows from twins by the automation and direction of data flow. Ebad *et al.* [2] reviewed 68 AI-driven mining DT studies. Birk *et al.* [6] present virtual sensors, with the KF

as an example, as a DT function and propose soft sensing of stockpile contents at a mine.

State estimation in mineral processing. Kalman-type estimators have been applied to a SAG mill [13] and to heap leaching, the latter evaluated in simulation only [14]; soft sensors in process industries are reviewed in [15]. The closest work is Meyer *et al.* [16], who applied an unscented KF with a nonlinear mass-conservation model, identified from production data, to a coal run-of-mine bin. Crushing-circuit models and controllers use mass balances [17], [18]; Itävuo *et al.* use sensor fusion to obtain the circuit inventory for control, but, in what we could read, the estimator and its consistency are not evaluated.

Consistency under mismatch. Statistical consistency tests are standard in tracking [19]; innovation-correlation tests and covariance identification go back to Mehra [20] and are surveyed in [21]. The effects of erroneous models [22], [23] and of colored measurement noise treated as white [24] are known. In the literature we reviewed, these tests have not been used to evaluate a hopper estimator under controlled process and sensor mismatch, which is the gap addressed here.

III. Methodology

A. Simulated process (ground truth)

With sampling period $\Delta t = 1\,\mathrm{s}$, the extraction capacity is $e_k^{\mathrm{cap}} = \max(\bar{e}+d_k, 0)$, where d_k is an Ornstein–Uhlenbeck deviation (time constant 600 s, stationary std. $0.08\,\mathrm{m^3/s}$) around $\bar{e} = 0.8\,\mathrm{m^3/s}$. Trucks arrive with gamma inter-arrival times (mean 210 s, coefficient of variation 0.4) and queue; a truck dumps $180\,\mathrm{m^3}$ in 45 s only if its load fits ($V_k + 180 \le V_{\max} = 450\,\mathrm{m^3}$). The effective extraction and the volume are

$$e_k = \min\!\big(e_k^{\mathrm{cap}},\ V_k/\Delta t + u_k\big), \tag{1}$$

$$V_{k+1} = V_k + \Delta t\,(u_k - e_k), \tag{2}$$

where u_k is the known dumping flow; (1) is the saturation that occurs when the hopper empties. All parameters are design assumptions of the order of magnitude of a large crushing station, not plant measurements.

B. Sensor

The measurement is $y_k = V_k + b_k + \sigma_k n_k$, where n_k is an AR(1) process with unit marginal variance and coefficient ρ, σ_k equals σ_v or $f_d\sigma_v$ during dumping, and $b_k = b_N k/N$ is a linear drift. Nominally, $\sigma_v = 10\,\mathrm{m^3}$ (an assumption reflecting the irregular surface of the rock pile, not catalog precision), $\rho = 0$, $f_d = 1$ and $b_N = 0$.

C. Estimator

The KF uses $\mathbf{x}_k = [V_k,\ e_k]^{\mathsf{T}}$,

$$A = \begin{bmatrix} 1 & -\Delta t \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} \Delta t \\ 0 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 \end{bmatrix}, \tag{3}$$

$Q = \mathrm{diag}(0, q_e)$ with $q_e = \sigma_e^2(1-a^2)$, $a = e^{-\Delta t/600}$ (the per-step variance of the hardness process, modeled by the filter as a random walk), $R = \sigma_v^2$, and a Joseph-form covariance update. The pair (A, H) is observable. The filter is not tuned on data, but its R and q_e equal the true sensor and smooth-process parameters, which is privileged information. It ignores mean reversion, saturation (1) and the bounds on V.

D. Steady-state characterization

For Δt short relative to the filter dynamics, the continuous-time counterpart ($\dot{V} = u - e$, $\dot{e} = w$, $y = V + v$) is a double integrator with the classical closed-form Riccati solution (second-order Butterworth poles; for the analogous α-β filter, [25] reports $\zeta \approx 0.7$). We use it to interpret the results, not as a contribution. With intensities $q = q_e/\Delta t$ and $r = R\,\Delta t$, the steady-state gain is $K = [\sqrt{2}\,\omega_n,\ -\omega_n^2]^{\mathsf{T}}$ and the error dynamics obey $s^2 + \sqrt{2}\,\omega_n s + \omega_n^2$: a second-order loop with

$$\zeta = 1/\sqrt{2}, \qquad \omega_n = \big(q_e/(R\,\Delta t^2)\big)^{1/4}. \tag{4}$$

The filter memory is about $\sqrt{2}/\omega_n$, only q_e/R matters (a common scaling of Q, R and P_0 leaves the gain unchanged, cf. [26]), and after a step Δe the remaining rate error is $\Delta e\, e^{-x}(\cos x + \sin x)$ with $x = \omega_n t/\sqrt{2}$, while the volume prediction bias peaks at $\Delta e\, e^{-\pi/4}/\omega_n$ about $1.11/\omega_n$ s after the step.

E. Baselines and metrics

All baselines are causal: raw measurement, moving average and local regression (LR, a least-squares line fitted to $z_k = y_k - \Delta t \sum_{j<k} u_j$ over the last W samples) for the volume; finite differences and the negative LR slope for the rate. LR and finite differences use u_k, as the KF does; the raw measurement and the moving average do not, so the KF's advantage over them includes the value of u_k. Each window is chosen from $W \in \{2, \ldots, 600\}$ by minimizing the true RMSE over 30 separate calibration runs, separately for volume and rate. For a comparison on equal footing, we also tune the KF on the same calibration runs by choosing the best factor on q_e from a grid, separately for volume and rate (E2b). Since only q_e/R affects the estimate, this single factor plays the role of W; keeping the true R only sets the scale of the NIS and NEES.

Per run, after a 300-step burn-in (the nominal gain is within 0.5 % of its steady value after step 202; with burn-ins of 60 or 600 steps the KF-vs-LR reductions change by about one percentage point at most and none changes sign): RMSE and MAE of V and e; NEES (expected 2; needs truth), NIS (expected 1) and the lag-1 autocorrelation r_1 of the normalized innovation (expected 0), the latter two computable in a plant [19], [20].

IV. Experimental Setup

Each condition uses $M = 100$ runs of one hour. Seeds derive from a master seed through NumPy's `SeedSequence` [27]. E1–E3b share one batch of trajectories and sensor draws, and E4 uses its own; within an experiment, comparisons are paired. We report the RMSE reduction $1 - \overline{\mathrm{RMSE}}_{\mathrm{KF}}/\overline{\mathrm{RMSE}}_{\mathrm{ref}}$ (positive favors the KF) with a paired bootstrap 95 % interval [28], the median per-run reduction,

the fraction of runs won and Wilcoxon tests [29] as descriptive statistics. After a first execution showed a larger error spread for the KF, we stratified runs by whether, and for how long, the hopper emptied; this stratification is exploratory. To test the explanation it suggested, we added an ablation (E2c): the same seeds with saturation disabled ($e_k = e_k^{\mathrm{cap}}$, V unbounded below), a physically impossible counterfactual. Experiments: E1, $\sigma_v \in \{2, 5, 10, 20, 40\}\,\mathrm{m}^3$ with $R = \sigma_v^2$ and re-tuned windows; E2, nominal; E3a, q_e multiplied by $f \in [0.01, 100]$ and a common scaling $c \in \{0.1, 1, 10\}$ of (q_e, R, P_0); E3b, ρ up to 0.95, f_d up to 5 and b_N up to $40\,\mathrm{m}^3$, one at a time, with nominal tuning for every estimator, plus $f \in \{0.1, 1, 10\}$ at $\rho = 0.95$ (also added after seeing the data); E4, a 25 % drop of $\bar{e}$ at $t = 1800\,\mathrm{s}$, with an NIS alarm (60-s causal mean above the 99th percentile of $\chi^2_{60}/60$) whose false-alarm rate for a perfectly consistent filter is estimated by Monte Carlo.

Before any experiment, the implementation is verified on data generated with the filter model (200 runs): symmetric positive-definite covariance, gain equal to an independent SciPy Riccati solution [30], unbiased error, NEES and NIS inside their 95 % χ^2 bands in 93.4 % and 94.9 % of the steps, white innovations, and identical online and batch results. The study runs in under a minute and two executions produce byte-identical result files. [Code availability: repository link or DOI to be added.]

V. Results

A. Nominal condition and the role of emptying

The hopper emptied in 34 of the 100 runs (1.9 % of the time). Fig. 1 shows the first run, which has two long emptying episodes; Table I summarizes E2. The KF reduced the volume RMSE by 74.7 % versus the raw measurement and the rate RMSE by 99.4 % versus finite differences. Against LR, the mean difference in volume was not significant (−3.4 % [−7.1, +0.8]), but the median per-run reduction was +8.1 % and the KF won 72 % of the runs. Without emptying, the KF won 98 % of the runs with a NEES of 1.94; with emptying it lost (−17.8 %) and its NEES rose to 36.1. The reduction decreased and changed sign as emptying lengthened, from +9.8 % (1–29 saturated steps, $n = 5$) to −25.9 % (≥120 steps, $n = 18$). In the ablation (E2c), the same 34 runs gave +9.6 % (97 % won) and a NEES of 1.95; runs without emptying are identical in both processes, and in 13 runs with emptying some truck dumps changed, because admission depends on V. With the calibrated tuning ($3q_e$ for the volume), the KF won 99 % of the runs (+5.3 %), with and without emptying, although only 4 of the 30 calibration runs had emptying. The steady-state error dynamics have $\omega_n = 0.02148$ rad/s and $\zeta = 0.7071$, matching (4) (0.02148), i.e., a memory of about 65.8 s.

B. Sensor noise

From $\sigma_v = 2$ to $40\,\mathrm{m}^3$ (E1), the mean volume reduction versus LR went from −14.1 % to +7.7 %, but this change came from the runs with emptying (−48.3 % to +9.6 %): without emptying the KF was better at every level, and the median per-run reduction did not grow with noise. The tuned LR windows grew from 45 to 300 s, and (4) predicts KF memories from 29 to 132 s.

TABLE I
Nominal condition (E2): mean RMSE over 100 runs and KF vs. LR comparison. ‡Same seeds, saturation disabled (ablation E2c). †Factor on q_e chosen on 30 calibration runs (3 for V, 1 for e).

Estimator	RMSE V [m^3]	RMSE e [m^3/s]
Raw measurement / FD	10.0	14.2
Moving average	5.45	—
Local regression	2.45	0.0828
Kalman filter	2.53	0.0812
KF vs. LR: RMSE reduction [%] (runs won [%])		
All runs, ratio of means ($n = 100$)	−3.4 (72)	+1.9 (84)
All runs, median per run	+8.1	+5.9
Runs without emptying ($n = 66$)	+10.4 (98)	+8.5 (100)
Runs with emptying ($n = 34$)	−17.8 (21)	−1.1 (53)
Same runs, saturation disabled‡	+9.6 (97)	+8.2 (97)
KF tuned on calibration runs†	+5.3 (99)	+1.9 (84)

C. Tuning

With $3q_e$ (Fig. 2) the KF beat LR on the volume in both regimes (+4.8 % without and +5.8 % with emptying) and on the rate in runs with emptying (+5.1 %); without emptying, the rate difference was not significant (+0.5 %, $p = 0.077$). Larger factors helped runs with emptying and hurt the others: $10q_e$ gave the lowest mean volume RMSE of the grid (2.29 vs. $2.32\,\mathrm{m}^3$ at $3q_e$) but a higher rate RMSE (0.0841 vs. $0.0799\,\mathrm{m}^3$/s), and the KF won only 29 % of the runs, losing every run without emptying. Increasing q_e monotonically reduced the NIS (it crossed 1 between $10q_e$ and $30q_e$) and r_1 (from 0.045 to 0.001 at $10q_e$). Scaling (q_e, R, P_0) by $c = 0.1$ left the RMSE unchanged but declared a variance ten times smaller than the nominal tuning (NEES 135.4 vs. 13.54).

D. Sensor mismatch and detectability

Table II summarizes E3b. With colored noise the KF's advantage grew (at $\rho = 0.95$ it won 97 % of the runs on the volume) and the NIS did not rise (0.80) while r_1 rose to 0.94 and the NEES to 41.1. At $\rho = 0.95$, raising q_e from 0.1 to 10 times its nominal value lowered the NIS from 1.065 to 0.605, barely changed r_1 (0.948 to 0.916) and increased both errors (volume 7.43 to $9.05\,\mathrm{m}^3$, rate 0.102 to $0.191\,\mathrm{m}^3$/s). At $\rho = 0.5$ the nominal NIS was 1.047 and r_1 0.515, yet raising q_e to $3q_e$ left the volume RMSE at $3.36\,\mathrm{m}^3$, raised the rate RMSE from 0.0874 to $0.0920\,\mathrm{m}^3$/s, and barely changed r_1 (0.557 to 0.486 from 0.1 to $10q_e$). With white noise the same change moved the NIS and r_1 toward 1 and 0 (r_1 from 0.122 to 0.001) and reduced the volume error (3.72 to $2.29\,\mathrm{m}^3$). With noise bursts the NIS rose to 5.80 and r_1 did not increase. With drift, NIS and r_1 equaled their nominal values while the NEES reached 385; the volume RMSE was $24.3\,\mathrm{m}^3$ for the KF and $24.2\,\mathrm{m}^3$ for LR, close to the $24.1\,\mathrm{m}^3$ RMS of the drift ramp over the evaluated window ($24.2\,\mathrm{m}^3$ adding the nominal KF error in quadrature).

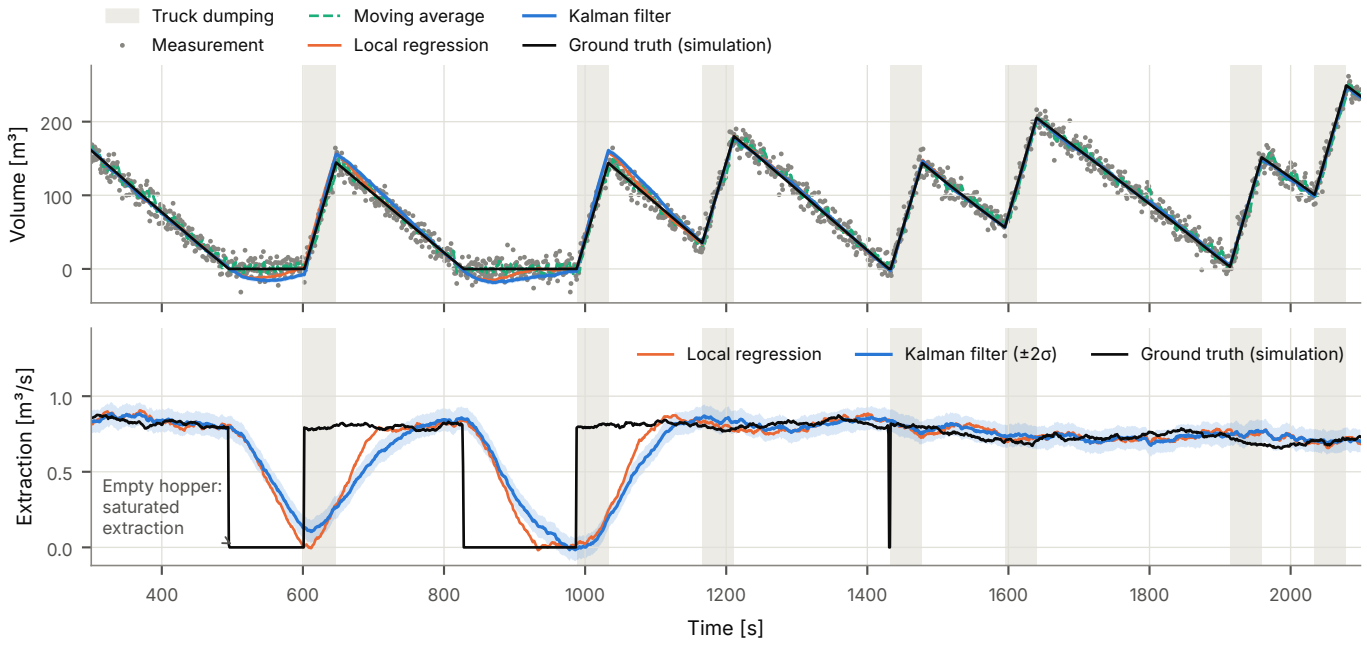


Fig. 1. First run (nominal condition, chosen by index). Top: volume; bottom: extraction rate with the KF $\pm 2\sigma$ band; shading marks truck dumping. During emptying the true extraction drops to zero; both estimators lag and the KF volume estimate becomes negative.

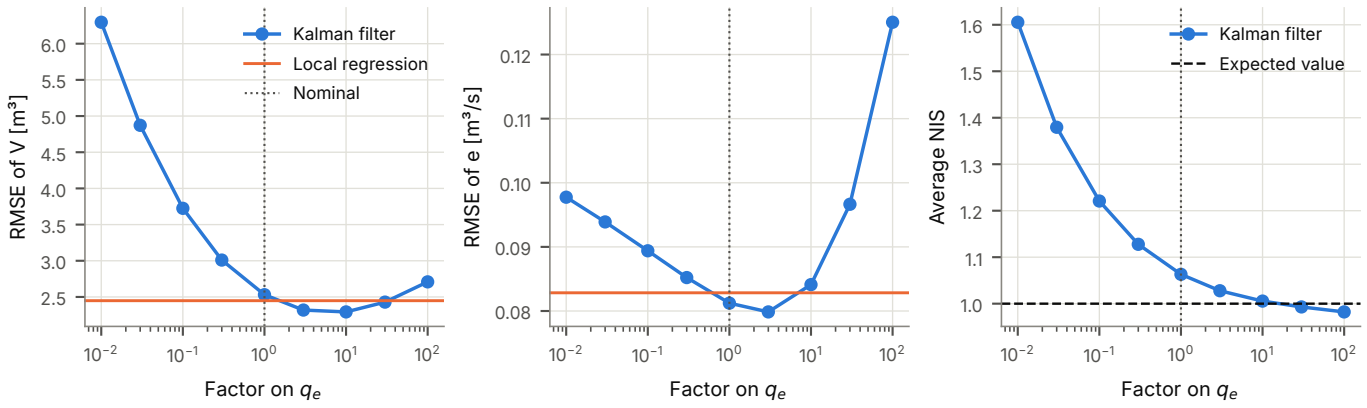


Fig. 2. Tuning error (E3a): KF volume and rate RMSE versus the factor on the assumed q_e (LR shown for reference) and mean NIS.

TABLE II
SENSOR MISMATCH (E3B): RMSE REDUCTION OF THE KF VS. LR (%), WITH THE RUNS WON (%) IN PARENTHESES AND THE MEDIAN PER-RUN REDUCTION, AND CONSISTENCY DIAGNOSTICS. NIS AND r_1 ARE COMPUTABLE IN A PLANT; NEES NEEDS THE GROUND TRUTH.

	KF vs. LR [%]			In-plant		Truth
Condition	V (wins)	V med.	e (wins)	NIS	r_1	NEES
Nominal	−3.4 (72)	+8.1	+1.9 (84)	1.06	0.05	13.5
$\rho = 0.9$	+12.5 (95)	+13.7	+16.1 (97)	0.93	0.89	28.9
$\rho = 0.95$	+11.8 (97)	+12.9	+17.8 (98)	0.80	0.94	41.1
$f_d = 5$	+9.4 (85)	+14.7	+10.6 (88)	5.80	0.01	18.5
$b_N = 40$ m^3	−0.1 (56)	0.0	+1.8 (84)	1.06	0.05	385

E. Abrupt hardness change

After the 25 % drop (E4, Fig. 3), the median rate error settled within ± 10 % of the step in 208 s for the KF and 96 s for LR; the KF absorbed 48 % of the step after 60 s and 88 % after 120 s (predicted: 44 and 89 %). It was worse than LR in the first five minutes (−1.9 %) and better before (+4.4 %) and after (+8.9 %). The across-run mean of the 60-s NIS rose from 1.025 to 1.157 at +82 s; the steady-state recursion predicts a peak of 1.158 at +86 s from a level of 1. After the change the mean extraction (0.6 m^3/s) falls well below the mean truck supply (0.857 m^3/s, vs. 0.8 m^3/s of extraction before), the hopper empties much less often (0.19 % vs. 0.88 % of steps) and the per-step NIS falls from 1.024 to 1.006; measured from that level, the rise is 0.15 (predicted: 0.16). The alarm fired in 29 % of runs after the change and 17 % before it; a perfectly consistent filter would give 21 % with no change at all.

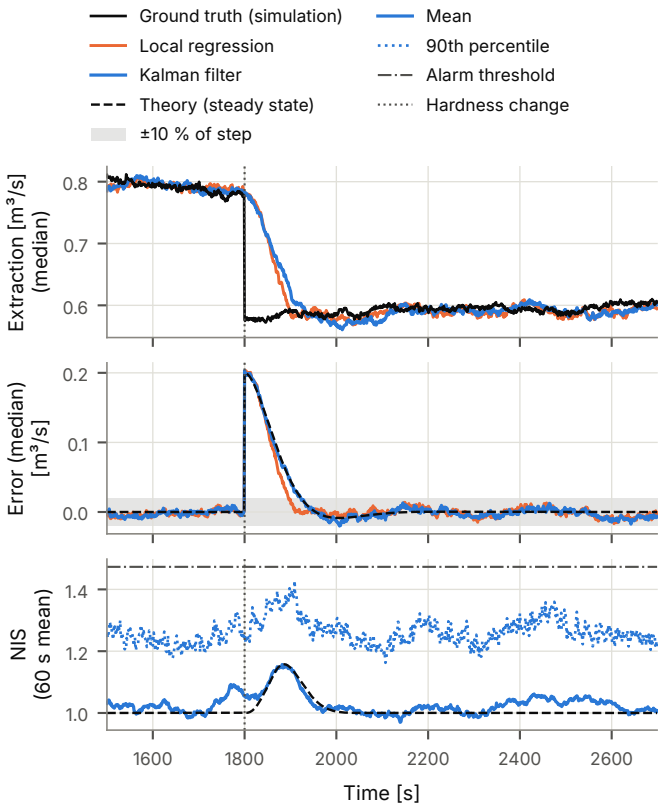


Fig. 3. Abrupt 25 % drop of the extraction capacity at $t = 1800$ s (E4), 100 runs: medians of true and estimated rate; median rate error with the steady-state prediction; mean 60-s NIS with its prediction and the alarm threshold.

VI. Discussion

Where the gain comes from. Most of the improvement over the raw measurement comes from structural information: the mass balance, on which mineral-processing observers are built [3], and the known dumping flow, both of which LR also exploits. The KF adds an exponential memory matched to q_e/R instead of a uniform window, which pays off when its model is approximately right, as in runs without emptying.

Saturation causes the losses. Saturation turns the extraction into a switching signal. The nominal KF, a $\zeta \approx 0.71$ loop with a memory of about a minute, keeps extrapolating an extraction that no longer occurs, driving its volume estimate below zero (minimum $-22.8\,\mathrm{m}^3$), whereas LR forgets the pre-change data after W samples; the longer the emptying, the larger the loss. A slightly larger q_e shortens the memory to about 50 s and compensates most of it at a small cost in normal operation; no single q_e is optimal for both regimes. With more sensor noise, LR also needs long windows and loses its advantage during emptying, which is why the mean benefit of the KF grows with noise. The ablation confirms the cause: with saturation disabled, the runs with emptying behave like the others, although mean reversion, the other nominal mismatch, remains.

What the in-plant diagnostics can see. Bursts inflate the innovation magnitude (NIS). Positively correlated noise is partly tracked as signal, so the innovation does not grow (the NIS fell from 1.06 to 0.80 as ρ rose to 0.95) while becoming correlated (r_1), and the true error exceeds the declared covariance [24]. For tuning, neither diagnostic suffices alone. An NIS above 1 can be removed by increasing q_e or R, which move the gain in opposite directions. Correlated innovations reveal a suboptimal filter [20], and with white noise a positive r_1 accompanied a low gain (E3a), but colored noise also produces it, and there raising the gain from its nominal value did not appreciably reduce the error. The NIS did not separate the two cases (at $\rho = 0.5$ it was 1.047); the response of r_1 to q_e did: it vanished with white noise and barely changed with colored noise. Since both diagnostics need no ground truth, this response can be checked in a plant by refiltering recorded data with another q_e. An r_1 that does not respond indicates colored noise but says nothing about the gain (at $\rho = 0.5$, going from 0.1 to $1\,q_e$ cut the volume RMSE from 4.12 to $3.36\,\mathrm{m}^3$ with r_1 almost unchanged); the noise should then be modeled before retuning the gain. Neither indicates the optimal value: r_1 vanished near $10q_e$, which has the lowest mean volume RMSE but not the lowest rate RMSE nor the best typical run. Drift is unobservable under the filter model: with $[V, e, b, \beta]$, $b_{k+1} = b_k + \Delta t \beta_k$ and $H = [1, 0, 1, 0]$, the observability matrix has rank 2 of 4. An abrupt change is absorbed progressively into the rate estimate, so the volume prediction bias peaks at only about 0.4 innovation standard deviations and the NIS barely reacts; the change is visible in the estimated rate itself.

Fairness. Neither side operates as in a plant: the baselines were tuned with ground truth, and the nominal KF uses the true noise parameters. The calibrated comparison (E2b) puts both on the same footing, one parameter tuned on the same runs, and favors the KF.

VII. Limitations

All evidence is from simulation with assumed parameters; it is not evidence of plant performance. The dumping flow is known without error; hardness dynamics, hopper geometry and sensor are simplified; there are no delays or data losses. The rate ground truth is the effective extraction, which is zero while the hopper is empty. The emptying stratification is exploratory and its strata are small; the ablation and the colored-noise tuning check were also decided after seeing the data; and many comparisons were made without multiplicity correction. The module has not been connected to a live data stream, so it is a digital model of a component, not a digital shadow or twin.

VIII. Conclusion

Under nominal conditions, a linear KF substantially improves on the raw hopper level, and against a window estimator that uses the same information it wins most runs; its losses concentrate in long emptying episodes, where it is also overconfident, an ablation shows that extraction saturation causes them, and a calibrated tuning largely removes them. Its self-assessment must be checked: the NIS detects noise bursts, the innovation autocorrelation detects colored noise but

indicates a low gain only if it responds to the tuning, and drift requires redundant measurements. Plant data, fusion with the downstream belt scale, and constraint-aware estimators are the next steps.

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[IEEE policy requires disclosing AI-generated content (text, figures, code) in this section, naming the system, the sections involved and the level of use. To be written and approved by the authors.]

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